

★ WEEK 11 · PROBLEM SET

Finance Applications: Three Hands-On Problems

Price the Free Lunch · Break Markowitz · Run HRP by Hand

Each problem: Statement → Principles & Lab Guide → Hand Calculation → Claude Code Prompt

Small numbers you can compute with a pencil — then a metaprompt that lets Claude Code scale the same idea to real data. Hand first, machine second.

Every Problem Follows the Same Four Steps

Work top to bottom — do not skip the hand calculation, it is where the understanding lives

1 · Statement

the question

A concrete finance scenario with a tiny dataset you can see at a glance.

one slide

2 · Principles

the tools

The formulas involved and a short lab guide: what to compute, in what order.

one slide

3 · Hand Calc

pencil first

Work every number by hand in a table.
Small n , clean arithmetic, no calculator fear.

1–2 slides

4 · Claude Code

scale it up

A metaprompt: paste it into Claude Code and it builds the full lab on real data.

one slide

Hand first, machine second — you check the machine, never the other way around.

Where the Data Comes From

Public / in-script sources first — the prompts fetch free data; CSVs are hand-calc references or offline fallbacks

Problem	Primary source (public · no signup)	Provided fallback file
P1 · Free lunch	numbers on the slides — pure pencil work	— (2-stock case, no file needed)
P2 · Markowitz flip	numbers on the slides + in-script check	— (2-asset case, no file needed)
P3 · HRP by hand	\$ 3b yfinance panel (10 tickers, free)	fml_w11_p3_clusters.csv (hand-calc)
Full-panel labs	same \$ 3b panel	fml_w11_panel_offline.csv (offline panel)

Course rule: if data is free online (yfinance, FRED, OpenML) or generated in-script, the prompt uses it.

P1

PROBLEM 1 • THE FREE LUNCH

Price the Free Lunch

Two stocks, three correlations — watch risk melt from 15% to 5%

Two Stocks, Fifty-Fifty, Three Worlds

$\sigma_1 = 20\%$, $\sigma_2 = 10\%$, $w = 50/50$ — compute σ_p for $\rho = +1, 0, -1$

Same two stocks, same weights, same expected returns in all three worlds.

Only the correlation ρ changes — and with it, the whole meaning of 'diversified'.

Given

- Individual vols: 20% and 10% (variances 400 and 100, in $\%^2$).
- Half the money in each stock.
- Three correlation worlds: +1, 0, -1.

Your deliverables

- σ_p^2 and σ_p for each ρ .
- Which world offers no lunch, some lunch, a feast — and why.
- One sentence: what Week 9's noisy correlations do to this menu.

The Tool: the Two-Stock Variance Formula

Three terms — two solo wobbles and one togetherness term that ρ controls

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \rho \sigma_1 \sigma_2 \quad (w_1=w_2=0.5, \sigma_1=20\%, \sigma_2=10\%)$$

Reading it

- Terms 1–2: each stock's own wobble, scaled by its weight squared.
- Term 3: the cross term — the only place ρ appears.
- ρ dials that term from +100 down to -100 (in $\%^2$) here.

Lab guide — per world

- Step 1: solo terms: $0.25 \cdot 400 + 0.25 \cdot 100 = 125$.
- Step 2: cross term: $2 \cdot 0.5 \cdot 0.5 \cdot \rho \cdot 20 \cdot 10 = 100\rho$.
- Step 3: $\sigma_p^2 = 125 + 100\rho$; take the square root.

Three Worlds Through One Formula

$125 + 100\rho$ — the whole story of diversification in one small sum

World	Cross term 100ρ	σp^2 (% ²)	σp	Lunch served?
$\rho = +1$ (clones)	+100	225	15.0%	none — one basket in disguise
$\rho = 0$ (strangers)	0	125	$\approx 11.2\%$	yes — below the 15% average
$\rho = -1$ (opposites)	-100	25	5.0%	a feast — risk nearly cancels

$$\rho = +1 : \sigma_p = 15\%$$

$$\rho = 0 : \sigma_p \approx 11.2\%$$

$$\rho = -1 : \sigma_p = 5\%$$

Correlation Is the Entire Menu

Nothing about the stocks changed — only how they dance together

15%

$\rho = +1$: the weighted average — zero diversification

11.2%

$\rho = 0$: independence alone buys a real discount

5%

$\rho = -1$: the umbrella-and-sunscreen feast

×3

Risk ratio between the worst and best world — same stocks

The Week 9 shadow

- The menu is priced off ρ — and measured ρ is mostly noise.
- A fake -0.5 promises a feast and delivers a stomachache.

Where P2 goes

- If risk depends this hard on ρ , what happens when an optimizer
- TRUSTS a noisy ρ of 0.95 completely? P2 detonates that landmine.

Scale It: the Lunch Menu on Real Pairs

Paste the metaprompt — Claude Code prices diversification across the whole panel

[INPUT] Prompt · Claude Code

```
Build a single Colab-ready Python script that:  
1) verifies the toy first: sigma (20%, 10%), w 50/50,  
   rho in [+1, 0, -1] -> confirm vols 15% / 11.18% / 5%;  
2) downloads the course panel with yfinance: tickers  
   ['AAPL', 'MSFT', 'GOOGL', 'AMZN', 'META', 'KO', 'PEP',  
    'XOM', 'JPM', 'JNJ'], 2020-2024; rets = pct_change();  
   (offline fallback: fml_w11_panel_offline.csv)  
3) for every pair of stocks: compute rho and the 50/50  
   portfolio vol vs the average of the two solo vols;  
   plot diversification benefit vs rho as a scatter;  
4) print the 3 best diversifying pairs and the 3 worst;  
5) three-sentence verdict: does the scatter follow the  
    $125 + 100 \cdot \rho$  logic, and which pair is the closest  
   thing to umbrella + sunscreen in this panel.
```

[OUTPUT] What to expect

```
toy check: 15 / 11.18 / 5  
           exactly  
scatter: benefit falls  
         linearly as rho rises  
best diversifiers: KO/PEP  
                  paired against tech  
worst: the tech twins  
  
Your 3-world table, priced  
on 45 real pairs.
```

P2

PROBLEM 2 · THE FRAGILE INVERSE

Break Markowitz on Purpose

Two near-twins, one tiny forecast change — a 53-point position swing

Two Near-Twin Stocks and a Forecast

Equal vols, correlation $\rho = 0.95$. Compute Markowitz weights $w \propto \Sigma^{-1}\mu$ for two forecasts

Case	Forecast μ_1 (stock 1)	Forecast μ_2 (stock 2)	What changed
A	10%	9.0%	—
B	10%	9.5%	μ_2 nudged by 0.5pp

Helper

- Equal vols: $w_1 : w_2 = (\mu_1 - \rho\mu_2) : (\mu_2 - \rho\mu_1)$.
- $0.95 \times 9.0 = 8.55 \cdot 0.95 \times 9.5 = 9.025$.

Your deliverables

- Normalized weights for both cases.
- The stock-2 swing, in points.
- IVP's answer on the same inputs.

The Tool: What Σ^{-1} Does to Near-Twins

The inverse divides by $(1 - \rho^2)$ — tiny for twins — and subtracts nearly-equal numbers

$$w \propto \Sigma^{-1}\mu \Rightarrow w_1 : w_2 = (\mu_1 - \rho \mu_2) : (\mu_2 - \rho \mu_1) \quad (\sigma_1=\sigma_2, \rho=0.95)$$

Why it explodes

- At $\rho = 0.95$ the twins are nearly the same bet — Σ is nearly singular.
- The weights ride on $\mu_1 - \rho \mu_2$: a DIFFERENCE of similar numbers.
- Small forecast noise $\rightarrow$ huge relative change in that difference.

Lab guide — per case

- Step 1: $u_1 = \mu_1 - 0.95 \cdot \mu_2, u_2 = \mu_2 - 0.95 \cdot \mu_1$.
- Step 2: normalize: $w = u / (u_1 + u_2)$.
- Step 3: compare the two cases; then note IVP's answer.

Two Forecasts, One Detonation

The arithmetic is trivial — the swing is not

	Case A: $\mu = (10\%, 9.0\%)$	Case B: $\mu = (10\%, 9.5\%)$
$u_1 = \mu_1 - 0.95 \mu_2$	$10 - 8.55 = 1.45$	$10 - 9.025 = 0.975$
$u_2 = \mu_2 - 0.95 \mu_1$	$9.0 - 9.5 = -0.50$	$9.5 - 9.5 = 0$
Weights (normalize)	(152.6%, -52.6%)	(100%, 0%)
IVP, same inputs	(50%, 50%)	(50%, 50%) — unmoved

$$\mu = (10\%, 9\%): w = (152.6\%, -52.6\%) \quad \mu = (10\%, 9.5\%): w = (100\%, 0\%)$$

A 0.5pp Forecast Moved a Position by 53 Points

No forecaster on earth is accurate to 0.5pp — so these weights are noise in a suit

53pp

Stock-2 position swing
from a 0.5pp forecast
tweak

−52.6%

Case A's short —
leverage
manufactured from
noise

$1-\rho^2$

= 0.0975: the tiny
denominator doing
the amplifying

0pp

IVP's swing on the
same inputs — the
meaning of robust

The general law

- Fragility peaks exactly where portfolios are largest: similar assets.
- The more twins in the universe, the wilder $\Sigma^{-1}\mu$ becomes.

What P3 offers

- IVP overreacts by ignoring ρ entirely.
- HRP re-admits correlation — but only through a tree, one fork at a time.

Scale It: Map the Fragility Surface

Paste the metaprompt — Claude Code shows exactly where Markowitz detonates

[INPUT] Prompt · Claude Code

Build a single Colab-ready Python script that:

- 1) verifies the toy first: equal vols, $\rho = 0.95$, $\mu = (10\%, 9\%)$ vs $(10\%, 9.5\%) \rightarrow$ confirm weights (152.6%, -52.6%) and (100%, 0%);
- 2) fragility surface: for ρ in $[0, 0.5, 0.9, 0.95, 0.99]$, recompute both cases and plot the stock-2 weight swing vs ρ ;
- 3) Monte Carlo: at $\rho = 0.95$, jitter both forecasts with 0.5pp Gaussian noise 1000 times; plot the histogram of the stock-2 weight; report its 5th and 95th percentiles;
- 4) run IVP on every draw and overlay its (constant) answer on the same histogram;
- 5) three-sentence verdict: where fragility explodes, and what the histogram says about trusting Sigma-inverse- μ weights.

[OUTPUT] What to expect

toy check: both cases exact
swing curve: hockey stick,
exploding past $\rho \sim 0.9$
histogram: stock-2 weight
smeared across $\pm 100\%$
IVP: one motionless bar
at 50%

Your 0.5pp detonation,
mapped 1,000 times.

P3

PROBLEM 3 · HRP

HRP by Hand — One Tree, Four Stocks

Within-cluster IVP, cluster variances, and the fair split across the fork

Four Stocks, Two Families

All variances = 1. Defensives {KO, PEP} have corr 0; tech twins {AAPL, AMZN} have corr 0.8. Allocate by HRP

Cluster	Members	Variances	Within-cluster corr
Defensive	KO, PEP	1, 1	0.0
Tech	AAPL, AMZN	1, 1	0.8

The HRP recipe

- Within each cluster: inverse-variance.
- Cluster portfolio variance $w'\Sigma w$.
- Across: weights $\propto 1/\text{cluster variance}$.

Your deliverables

- Within weights, both cluster variances.
- The across split; final four weights.
- Why HRP disagrees with $25\% \times 4$.

The Tool: Let the Cluster's Own Risk Set Its Share

Correlated members make a cluster riskier as a unit — the fork notices

$$\text{within: } w_i \propto 1/\sigma_i^2 \quad \text{across: } w_{\text{cluster}} \propto 1/\sigma_{\text{cluster}}^2 \quad \sigma_{\text{cluster}}^2 = w^\top \Sigma_{\text{cluster}} w$$

Reading it

- Equal variances $\rightarrow$ within-cluster IVP gives 50/50 in both families.
- Cluster variance = $0.25\sigma_1^2 + 0.25\sigma_2^2 + 2(0.25)\rho\sigma_1\sigma_2$.
- Correlation only enters HERE — locally, inside its own branch.

Lab guide

- Step 1: within weights (50/50 each).
- Step 2: cluster variances: 0.5 (defensive) vs 0.9 (tech).
- Step 3: split $\propto (1/0.5, 1/0.9)$; multiply back down to stocks.

Down the Tree in Three Rows

The correlated twins get taxed for being one bet

Step	Defensive {KO, PEP}	Tech {AAPL, AMZN}
Within (IVP)	50 / 50	50 / 50
Cluster variance	$0.25 + 0.25 + 0 = 0.5$	$0.25 + 0.25 + 2(0.25)(0.8) = 0.9$
Across split $\propto 1/\text{var}$	$2 \rightarrow 2/(2+1.11) = 64.3\%$	$1.11 \rightarrow 35.7\%$
Final weights	KO 32.1% · PEP 32.1%	AAPL 17.9% · AMZN 17.9%

$$\sigma_D^2 = 0.5, \quad \sigma_T^2 = 0.9 \Rightarrow 64.3\% / 35.7\% \Rightarrow w = (32.1, 32.1, 17.9, 17.9)\%$$

64.3 / 35.7 — the Tree Taxed the Twins

Correlation-blind methods say 25% each; HRP sees two bets and a half, not four

0.9 vs 0.5

Cluster variances —
the tech pair is riskier
AS A UNIT

64.3%

Share to the true
diversifiers
(defensives)

32.1%

Each defensive stock
— up from a naive
25%

0

Matrix inversions
performed anywhere
in the process

The local-damage property

- A noisy tech correlation only distorts the tech branch.
- In Markowitz it would poison all four weights at once.

Scaling to NCO

- Replace 'IVP within' by 'optimize within the small cluster' → NCO.
- Same tree, more machinery — still no giant inversion.

Scale It: Full HRP on the Real Panel

Paste the metaprompt — Claude Code grows the tree and races the allocators

[INPUT] Prompt · Claude Code

```
Build a single Colab-ready Python script that:
1) verifies the toy first: 4 stocks, variances 1,
   clusters (corr 0) and (corr 0.8) -> confirm HRP
   weights (32.1, 32.1, 17.9, 17.9)%;
   (hand-calc reference: fml_w11_p3_clusters.csv)
2) downloads the course panel with yfinance (10
   tickers, 2020-2024); rets = pct_change();
   (offline fallback: fml_w11_panel_offline.csv)
3) implements full HRP: correlation distance, single
   linkage, quasi-diagonal order, recursive bisection
   with cluster variance; plot the dendrogram and
   print the weights;
4) races 1/N, IVP, HRP, raw min-variance: in-sample
   vol + max/min weight + bootstrap weight turnover
   (20 resamples);
5) three-sentence verdict in this week's language.
```

[OUTPUT] What to expect

```
toy check: 32.1 / 17.9
           exactly
dendrogram: defensives and
           tech in separate branches
min-var: extreme weights,
           high turnover
HRP: gentle weights, low
           turnover

Your one-fork tree, grown
to ten branches.
```

Problem Set — Three Things Your Hands Now Know

You computed every number yourself; the code only repeats what you already understand

1

The free lunch is $125 + 100\rho$: vol 15% $\rightarrow$ 11.2% $\rightarrow$ 5%

Same stocks, same weights — correlation alone set the menu, and measured ρ is mostly noise (W9).

2

Markowitz flipped 53 points on a 0.5pp forecast

Near-twins make Σ^{-1} explosive: differences of similar numbers over $(1 - \rho^2)$. IVP didn't move at all.

3

HRP taxed the twins: 64.3 / 35.7, zero inversions

Within-IVP, cluster variances 0.5 vs 0.9, a fair fork — correlation admitted locally, never amplified globally.

Hand calculation first — then let Claude Code scale the same idea to real data.